

HDOC Day-9 Assignment

Question 1; Problem Statement:

Write a menu-driven C program to input two numbers and swap them using a third variable and without using a third variable. Also display the values before and after swapping.

1. Algorithm

Option 1: Swap using a third variable

1. Start.
2. Read two numbers a and b.
3. Print original values of a and b.
4. Store value of a in a temporary variable temp (temp=a).
5. Assign value of b to a (a=b).
6. Assign value of temp to b (b=temp).
7. Print updated values of a and b.
8. Stop.

Option 2: Swap without using a third variable

1. Start.
2. Read two numbers a and b.
3. Print original values of a and b.
4. Set $a=a+b$.
5. Set $b=a-b$.
6. Set $a=a-b$.
7. Print updated values of a and b.
8. Stop.

2. Pseudocode

BEGIN

PRINT "Menu:"

PRINT "1. Swap using third variable"

PRINT "2. Swap without using third variable"

PRINT "Enter choice: "

READ choice

IF choice == 1 THEN

PRINT "Enter two numbers: "

READ a, b

PRINT "Before Swapping: a = ", a, ", b = ", b

temp = a

a = b

b = temp

PRINT "After Swapping: a = ", a, ", b = ", b

ELSE IF choice == 2 THEN

PRINT "Enter two numbers: "

READ a, b

PRINT "Before Swapping: a = ", a, ", b = ", b

a = a + b

b = a - b

a = a - b

PRINT "After Swapping: a = ", a, ", b = ", b

ELSE

PRINT "Invalid Choice"

ENDIF

END

3. Test Case Table

TEST CASE	CHOICE	INPUT 1 (a)	Input 2 (b)	Expected output	status
1.	1	10	20	Before: a=10,b=20 After: a=20,b=10	pass
2.	2	5	15	Before: a=5,b=15 After: a=15,b=5	pass
3.	3	na	na	Invalid choice!	pass

5. C Program Code

UW PICO 5.09

```
#include <stdio.h>

int main() {
    int choice;
    int a, b, temp;

    printf("--- MENU ---\n");
    printf("1. Swap using third variable\n");
    printf("2. Swap without third variable\n");
    printf("Enter your choice (1 or 2): ");
    scanf("%d", &choice);

    switch(choice) {
        case 1:
            printf("Enter first number (a): ");
            scanf("%d", &a);
            printf("Enter second number (b): ");
            scanf("%d", &b);

            printf("\nBefore Swapping: a = %d, b = %d\n", a, b);

            temp = a;
            a = b;
            b = temp;

            printf("After Swapping: a = %d, b = %d\n", a, b);
            break;

        case 2:
            printf("Enter first number (a): ");
            scanf("%d", &a);
            printf("Enter second number (b): ");
            scanf("%d", &b);

            printf("\nBefore Swapping: a = %d, b = %d\n", a, b);

            a = a + b;
            b = a - b;
            a = a - b;
```

```

        printf("After Swapping: a = %d, b = %d\n", a, b);
        break;

    default:
        printf("\nInvalid Choice! Please select 1 or 2.\n");
    }

    return 0;
}

```

- EXECUTION AND CODE OUTPUT :

```

[manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % gcc q1.c -o q1
[manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % ./q1
--- MENU ---
1. Swap using third variable
2. Swap without third variable
Enter your choice (1 or 2): 1
Enter first number (a): 10
Enter second number (b): 20

Before Swapping: a = 10, b = 20
After Swapping: a = 20, b = 10
manalithakar@Unknown_8e:80:9d:9c:62:11 ~ %

```

Question 2

Problem Statement:

Write a menu-driven C program to find the sum of the first n natural numbers and sum of squares of the first n natural numbers.

1. Algorithm

Option 1: Sum of first n natural numbers

1. Start.
2. Input value of n.
3. Initialize sum = 0.
4. Loop i from 1 to n:
 - a. $\text{sum} = \text{sum} + i$
5. Print sum.
6. Stop.

Option 2: Sum of squares of first n natural numbers

1. Start.
2. Input value of n.
3. Initialize sum_sq = 0.
4. Loop i from 1 to n:
 - a. $\text{sum_sq} = \text{sum_sq} + (i * i)$
5. Print sum_sq.
6. Stop.

2. Pseudocode

BEGIN

PRINT "Menu:"

PRINT "1. Sum of first n natural numbers"

```
PRINT "2. Sum of squares of first n natural numbers"
```

```
PRINT "Enter choice: "
```

```
READ choice
```

```
IF choice == 1 THEN
```

```
    PRINT "Enter n: "
```

```
    READ n
```

```
    sum = 0
```

```
    FOR i = 1 TO n DO
```

```
        sum = sum + i
```

```
    ENDFOR
```

```
    PRINT "Sum = ", sum
```

```
ELSE IF choice == 2 THEN
```

```
    PRINT "Enter n: "
```

```
    READ n
```

```
    sum_sq = 0
```

```
    FOR i = 1 TO n DO
```

```
        sum_sq = sum_sq + (i * i)
```

```
    ENDFOR
```

```
    PRINT "Sum of squares = ", sum_sq
```

```
ELSE
```

```
    PRINT "Invalid Choice"
```

```
ENDIF
```

END

4. Test Case Table

Test case	choice	Input(n)	Expected op	status
1.	1	5	Sum = 15	pass
2.	2	3	Sum of sq=14	pass

5. C Program Code

```
#include <stdio.h>

int main() {
    int choice, n, i;
    int sum = 0, sum_sq = 0;

    printf("--- MENU ---\n");
    printf("1. Sum of first n natural numbers\n");
    printf("2. Sum of squares of first n natural numbers\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch(choice) {
        case 1:
            printf("Enter the value of n: ");
            scanf("%d", &n);

            for(i = 1; i <= n; i++) {
                sum = sum + i;
            }

            printf("Sum of first %d natural numbers is: %d\n", n, sum);
            break;

        case 2:
            printf("Enter the value of n: ");
            scanf("%d", &n);

            for(i = 1; i <= n; i++) {
                sum_sq = sum_sq + (i * i);
            }

            printf("Sum of squares of first %d natural numbers is: %d\n", n, sum_sq);
            break;

        default:
            printf("Invalid Choice!\n");
    }
}
```

```
return 0;
}
```

^G Get Help ^O WriteOut ^R Read File ^Y
^X Exit ^J Justify ^W Where is ^V

- EXECUTION AND CODE OUTPUT :

```
Terminal — zsh — 182x44

[manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % nano q2.c
[manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % gcc q2.c -o q2
[manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % ./q2
--- MENU ---
1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
Enter your choice: 1
Enter the value of n: 5
Sum of first 5 natural numbers is: 15
[manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % ./q2
--- MENU ---
1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
Enter your choice: 2
Enter the value of n: 3
Sum of squares of first 3 natural numbers is: 14
manalithakar@Unknown_8e:80:9d:9c:62:11 ~ %
```

Question 3

Problem Statement:

Write a menu-driven C program to calculate simple interest, compound amount, compound interest, and total payable amount for given principal, rate, and time.

1. Algorithm

1. Start.
2. Read principal, rate, and time.
3. Display menu with options:
 - a. Option 1: Calculate Simple Interest & Total Amount
 - b. Option 2: Calculate Compound Amount & Compound Interest
4. Input choice.
5. If choice == 1:
 - a. $SI = \text{principal} * \text{rate} * \text{time} / 100$
 - b. $\text{Total Amount} = \text{principal} + SI$
 - c. Display SI and Total Amount.
6. If choice == 2:
 - a. $\text{Compound Amount} = \text{principal} * (1 + \text{rate} / 100)^{\text{time}}$
 - b. $\text{Compound Interest} = \text{Compound Amount} - \text{principal}$
 - c. Display Compound Amount and Compound Interest.
7. Else:
 - a. Print "Invalid Choice".
8. Stop.

2. Pseudocode

```
BEGIN PRINT "Enter Principal: " READ p PRINT "Enter Rate of Interest: " READ r PRINT  
"Enter Time in years: " READ t
```

```
PRINT "Menu:"
```

```
PRINT "1. Simple Interest and Total Amount"
```

```
PRINT "2. Compound Amount and Compound Interest"
```

```
READ choice
```

```
IF choice == 1 THEN
```

```

    si = (p * r * t) / 100
    total = p + si
    PRINT "Simple Interest = ", si
    PRINT "Total Payable Amount = ", total
ELSE IF choice == 2 THEN
    amount = p * pow((1 + r / 100), t)
    ci = amount - p
    PRINT "Compound Amount = ", amount
    PRINT "Compound Interest = ", ci
ELSE
    PRINT "Invalid Choice"
ENDIF

END

```

3. Test Case Table

Test case	Principal(p)	Rate (r)	Time (t)	choice	Expected op	status
1.	1000	5	2	1	SI = 100.00 Total Amount = 1100.00	pass
2.	1000	5	2	2	Compound Amount = 1102.50 CI = 102.50	pass
3.	500	2	1	4	Invalid choice!	pass

5. C Program Code

```
UW PICO 5.09 File: q3.c

#include <stdio.h>
#include <math.h>

int main() {
    float principal, rate, time;
    float si, total_amount, compound_amount, ci;
    int choice;

    printf("Enter Principal amount: ");
    scanf("%f", &principal);

    printf("Enter Rate of interest: ");
    scanf("%f", &rate);

    printf("Enter Time period (in years): ");
    scanf("%f", &time);

    printf("\n--- MENU ---\n");
    printf("1. Calculate Simple Interest and Total Amount\n");
    printf("2. Calculate Compound Amount and Compound Interest\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch(choice) {
        case 1:
            si = (principal * rate * time) / 100.0;
            total_amount = principal + si;

            printf("\nSimple Interest = %.2f\n", si);
            printf("Total Payable Amount = %.2f\n", total_amount);
            break;

        case 2:
            compound_amount = principal * pow((1 + rate / 100.0), time);
            ci = compound_amount - principal;

            printf("\nCompound Amount = %.2f\n", compound_amount);
            printf("Compound Interest = %.2f\n", ci);
            break;

        default:
            printf("\nInvalid Choice!\n");
    }

    return 0;
}
```

- EXECUTION AND CODE OUTPUT :

```
manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % ./q3
Enter Principal amount: 1000
Enter Rate of interest: 5
Enter Time period (in years): 2

--- MENU ---
1. Calculate Simple Interest and Total Amount
2. Calculate Compound Amount and Compound Interest
Enter your choice: 1

Simple Interest = 100.00
Total Payable Amount = 1100.00
manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % ./q3
Enter Principal amount: 1000
Enter Rate of interest: 5
Enter Time period (in years): 2

--- MENU ---
1. Calculate Simple Interest and Total Amount
2. Calculate Compound Amount and Compound Interest
Enter your choice: 2

Compound Amount = 1102.50
Compound Interest = 102.50
manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % █
```